Lecture 6.2: Structural Models

AREC 570 / ECON 530 · Unit 6

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Before class

Assigned

- Nevo and Whinston, “Taking the Dogma out of Econometrics: Structural Modeling and Credible Inference” (Nevo and Whinston 2010)
- Nevo, “A Practitioner’s Guide to Estimation of Random-Coefficients Logit Models of Demand” (optional) (Nevo 2000)

Readings and course handouts are available on Canvas.

Also used in these slides

- Timmins and Schlenker, from Lecture 6.1 (Timmins and Schlenker 2009)
- Bellemare, on notation (Bellemare 2020)
- BLP demand: Berry (1994); Berry et al. (1995); Nevo (2001)
- Conlon and Gortmaker, on best practices and PyBLP (Conlon and Gortmaker 2020)

Latin letters mark what you observe; Greek letters mark what you estimate

	Latin letters	Greek letters
Examples	y, x, W, P, V	α alpha, β beta, γ gamma, δ delta, ε epsilon
Used for	Variables you observe in the data	Parameters you estimate, and the error term you never observe

- Latin letters are the letters of the English alphabet
- Fixed effects are parameters, so they get Greek letters: γ_c, δ_t
- A Greek letter tells the reader the term is estimated

Source: Bellemare (2020), Section 5.1.

Subscripts tell the reader the level at which each term varies

$$y_{cst} = \beta P_{st} + \gamma_c + \delta_t + \varepsilon_{cst}$$

- One letter per dimension, used the same way throughout the paper: c county, s state, t year
- Order from the smallest unit to the largest, with time last
- A term carries only the subscripts it varies over: a state policy is P_{st}
- A fixed effect's subscript says what it holds constant: γ_c county, δ_t year
- The outcome's subscripts state the unit of observation: a county in a year

Reduced-form and structural approaches

A structural model recovers the parameters of behavior

- Timmins and Schlenker's definition: any empirical model that recovers the parameters of behavioral functions, such as demand and supply equations, utility functions, or cost structures
- It starts from a theoretical model and states what the researcher can and cannot observe
- What cannot be observed becomes a structural error, backed out of observed behavior
- The parameters are estimated with a procedure such as GMM or maximum likelihood

Source: Timmins and Schlenker (2009), Section 2.

The term “reduced form” has two meanings

- **Textbook meaning:** solve a system of equations so the outcome depends only on exogenous variables
- **Recent meaning:** program evaluation or treatment effects, with attention to how the exogenous variable was assigned
- The goal is to avoid as many structural assumptions as possible
- Reduced-form studies still require assumptions of their own

The designs from Lecture 6.1 are reduced form in the recent sense: fixed effects, quasi-experiments, regression discontinuity, and randomization.

Source: Timmins and Schlenker (2009), Section 3.

The two approaches differ in what they estimate and what they assume

	Reduced form	Structural
Estimates	High-level response parameters, such as the effect of a policy	Behavioral parameters, such as preferences and costs
Credibility comes from	The source of variation	The source of variation and the economic model
Best suited to	Establishing causality for a change in the data	Feedback loops, large policy changes, and welfare
Main assumptions	How treatment was assigned	Functional form, the distribution of unobservables, and behavior
Main limit	Harder to extend to changes not observed	Conclusions depend on the model being right

Source: Timmins and Schlenker (2009), Sections 1 and 4.

Structural models are used for feedback effects, new policies, and welfare

- **Feedback loops:** people and firms re-optimize, and markets re-equilibrate
- **Large or new policies:** the change has not happened, or not at this scale
- **Welfare:** effects are converted to dollars through a utility function
- **Who bears the cost:** heterogeneity in preferences changes the distributional answer
- **Unintended consequences:** simulate the world with one piece of the policy turned off

Source: Timmins and Schlenker (2009), Section 2.1.

Structural analysis recovers parameters, then simulates counterfactuals

1. Match observed behavior to a theoretical model to recover parameters such as preferences and technology
2. Use the model to predict responses to changes, assuming those parameters are unchanged

Nevo and Whinston describe structural analysis and credible identification as complements: identification establishes what the data reveal, and the model extends those results to settings not observed. Timmins and Schlenker conclude that the most convincing evidence comes from contrasting approaches.

Sources: Nevo and Whinston (2010), pages 70 to 71; Timmins and Schlenker (2009), Section 4.

Merger analysis shows what each approach can and cannot answer

	Treatment effect of a past merger	Merger simulation
Method	Compare prices near and far from the merging firms, before and after	Estimate demand and pricing, then simulate joint pricing
Strength	Clear identification for that merger	Predicts a proposed merger, with welfare
Limit	Other mergers may not be similar; no welfare measure	Holds products, firms, and pricing behavior fixed; instruments are often imperfect

Nevo and Whinston's conclusion: each is useful in some cases and fails in others, depending on the question and the data.

Source: Nevo and Whinston (2010), pages 73 to 77.

Types of structural models you may see

Topic	Structural models	What they recover
Consumer demand	Demand systems (AIDS, QUAIDS); discrete choice (logit to BLP)	Elasticities, substitution, welfare
Nonmarket values	Hedonic; Ricardian; travel cost; choice experiments	Willingness to pay
Producers	Production, cost, and profit functions; stochastic frontier	Returns to scale, efficiency
Firms and markets	Conduct and markups; equilibrium displacement	Markups, pass-through, incidence
Dynamics and risk	Real options; dynamic discrete choice; risk preferences and insurance demand	Value of waiting, risk aversion
Trade and policy	Structural gravity; spatial equilibrium; CGE	Trade costs, economy-wide effects

Each method has an example specification on the methods map.

BLP: one structural model

A structural section states the same four choices as any empirical section

Choice	In a structural paper
Model	The primitives: preferences, technology, and how agents interact
Estimation and inference	The estimator in one paragraph, with details in an appendix
Identification	Which variation identifies which parameter
Reported results	Fit or validation, and the counterfactuals the model is used for

The order matches Lecture 6.1. The methods map has the details, including the steps of BLP estimation.

BLP addresses the large number of cross-price effects in differentiated-product demand

- With 100 products, a log-log demand system has at least 10,000 parameters
- BLP describes each product by its characteristics, so the number of parameters depends on the number of characteristics
- It uses market-level data on prices and quantities, and it allows for endogenous prices
- **Running question:** how much would prices rise if two cereal firms merged?

Sources: Nevo (2000), pages 514 to 515; Berry et al. (1995).

Model: each consumer chooses the product that gives the highest utility

$$u_{ijt} = \alpha_i(y_i - p_{jt}) + x_{jt}\beta_i + \xi_{jt} + \varepsilon_{ijt}$$

- u_{ijt}: utility of consumer i from product j in market t (a city in a quarter)
- y_i: income; p_{jt}: price; x_{jt}: observed characteristics, such as sugar content
- ξ_{jt}: characteristics the researcher does not observe, such as brand quality or promotions
- α_i, β_i: tastes that vary across consumers with demographics and unobserved traits
- ε_{ijt}: a mean-zero shock; consumers can also buy nothing (the outside good)

Summing individual choices gives each product's predicted market share. The data are the observed shares.

Source: Nevo (2000), pages 517 to 520.

Estimation: invert shares to recover the unobserved term, then use GMM

In plain logit, where tastes do not vary, shares invert to a linear regression:

$$\ln s_{jt} - \ln s_{0t} = x_{jt}\beta - \alpha p_{jt} + \xi_{jt}$$

- s_{jt}: market share of product j; s_{0t}: share of the outside good
- This is a linear instrumental-variables regression (Berry 1994)
- With random coefficients, the same idea is solved numerically for each guess of the taste parameters
- The estimator is GMM: instruments must be uncorrelated with ξ_{jt}

In a paper, this gets one paragraph. The steps go in an appendix.

The appendix reports how the model was solved

Same model, same data	Mean price coefficient	Mean own-price elasticity
Replication of the published estimates	−32.4	−3.70
Tighter optimizer tolerance	−62.7	−3.62
Best practices	−27.5	−3.69

- Report the optimizer, tolerances, starting values, and integration method
- Try several optimizers and starting values, and check that they agree
- Report the quantities that matter for the question, because individual coefficients can move a lot

Source: Conlon and Gortmaker (2020), Table 7.

Identification of the price coefficient: price is endogenous, so it needs instruments

- Price is an equilibrium outcome: firms set markups knowing demand, so p_{jt} is correlated with ξ_{jt}
- OLS biases the price coefficient toward zero, because high-ξ_{jt} products have both higher prices and higher shares
- Identification requires instruments Z_{jt} that shift price and satisfy $E[\xi_{jt} \mid Z_{jt}] = 0$

Instrument	Why it moves price	Exclusion restriction
Cost shifters	Shift marginal cost	Uncorrelated with demand shocks
Characteristics of rival products (BLP)	Close rivals lower markups	Product characteristics are set before demand shocks are realized
Prices of the same brand in other markets (Hausman)	Same brand shares marginal cost across markets	Demand shocks are independent across markets

Source: Nevo (2000), pages 528 and 534 to 536.

Identification of substitution patterns: taste parameters need variation across markets

- Plain logit implies substitution proportional to market shares (independence of irrelevant alternatives)
- Random coefficients let substitution depend on product characteristics
- The taste distribution is identified by how shares respond across markets with different choice sets, characteristics, and prices
- Parameters linking tastes to demographics are identified by variation in the distribution of demographics across markets
- Identification improves with more markets and more variation in the set of products

Source: Nevo (2000), pages 523 and 529 to 530.

Reported results: state what the counterfactual holds fixed, and how you validated the model

- Merger simulation: set the prices of the merging firms jointly and solve for new prices
- Report results under a range of assumptions
- State what is held fixed: current merger simulations hold products, firms, and pricing behavior fixed
- Validate the model: check that it predicts outcomes not used to estimate it

Sources: Nevo and Whinston (2010), pages 75 to 77; Timmins and Schlenker (2009), Section 2.

Demo: cereal results

Demo: estimating cereal demand in steps, then simulating a merger

- **Data:** Nevo's simulated cereal data, 24 products in 94 markets (47 cities, 2 quarters)
- Nevo: the data are "not real and should not be used for any analysis," so the numbers here illustrate the method
- **Steps:** plain logit by OLS, add product fixed effects, add instruments, add random coefficients, simulate a merger of firms 1 and 2
- **Software:** PyBLP in Python; the script runs in under a minute

Open the cereal demand demo

Sources: Nevo (2000), page 538; Conlon and Gortmaker (2020).

Product fixed effects triple the price coefficient, because quality was omitted

Specification (plain logit)	Price coefficient
OLS, no product fixed effects	−10.1
OLS, product fixed effects	−29.0
Instrumental variables, product fixed effects	−30.0

- Better cereals cost more, so without fixed effects a high price partly signals high quality
- Product fixed effects absorb a product's fixed quality, which is part of ξ_{jt}
- In these simulated data, the instruments change little once fixed effects are included

Two models with the same own-price elasticity give different merger predictions

	Plain logit	Random coefficients
Median own-price elasticity	−3.65	−3.61
Largest rival response ÷ smallest	1.0	6.2
Median price change, merging firms	5.8%	11.6%
Median price change, other firms	0.1%	0.4%
Change in consumer surplus	−11.6%	−13.6%

The counterfactual depends on how the model lets consumers substitute, which is an assumption the section has to state and defend.

Using a structural model

A structural section reports five items

1. The model's primitives
2. Which variation identifies which parameter
3. The estimator in one paragraph, with details in an appendix
4. How well the model fits or validates
5. The counterfactuals it is used for

This is row 10 of the empirical strategy checklist.

Pairing reduced-form evidence with a structural model is good practice

- A credible design shows that the response exists in the data
- Reduced-form estimates can serve as inputs to the model or as targets it must reproduce
- Retrospective studies check the accuracy of model predictions
- Reporting why the approaches agree or disagree gives the reader more confidence than either one alone

Sources: Timmins and Schlenker (2009), Sections 1 and 4; Nevo and Whinston (2010), page 77.

What to think about when using a structural model

- **The question:** does it require a model? Name the reason: feedback, a new policy, welfare, or distribution
- **The parameters:** which ones does the model recover, and what variation identifies each one?
- **The assumptions:** which ones drive the counterfactual, and how do results change under alternatives?
- **The reduced-form evidence:** what credible estimate supports the model?
- **The validation:** does the model predict outcomes not used to estimate it?

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Course navigation

- All lectures
- Schedule and preparation
- Course reading list
- Empirical strategy checklist
- Methods map
- Cereal demand demo
- Download this lecture as a PDF